

Mathematics for Data Science

Exercise Sheet – Lecture 2

Matrices of a linear application and change of basis

M1 – AI / Data Science

Fall 2026

Unless stated otherwise, all vector spaces are real vector spaces.

Part 1 – Course questions

Part 1(a) – Short exercises

Exercise 1

Find the matrix corresponding to the linear transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

defined by:

$$T(e_1) = (1, 0, 0)^T, \quad T(e_2) = (0, 1, 0)^T, \quad T(e_3) = (0, 0, 1)^T$$

where $\{e_1 = (1, 0, 0)^T, e_2 = (0, 1, 0)^T, e_3 = (0, 0, 1)^T\}$ (the standard basis).

Exercise 2

Consider the linear transformation

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T(x_1, x_2) = (x_1, 2x_2).$$

1 - Give the matrix representation of T in the standard basis.

2 - Consider now the basis

$$B' = (v_1, v_2), \quad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

What is the matrix representation of T in this basis?

Exercise 3

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Verify directly that the (i, j) entry of AB is:

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Exercise 4

Compute the rank and kernel of the matrix, then determine $\text{rank}(A)$ and $\dim(\ker(A))$.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Exercise 5

Are the following matrices invertible? If so, compute their inverse.

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix};$$

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}; C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}; D = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Exercise 6

Let $A \in \mathbb{R}^{n \times n}$. Prove that the following statements are equivalent:

1. The transformation $x \mapsto Ax$ is bijective.
2. $\text{Im}(A) = \mathbb{R}^n$.
3. $\ker(A) = \{0\}$.

Exercise 7 (*)

Find some examples of 3×3 orthogonal matrices.

Part 1(b) – True/False questions

For each statement, decide whether it is true or false, and justify your answer.

T/F 1. A matrix $A \in \mathbb{R}^{m \times n}$ is invertible if and only if $m = n$.

T/F 2. If A is a diagonal matrix, then A^2 is also diagonal.

T/F 3. If A^2 is diagonal, then A is also diagonal.

T/F 4. If a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is injective, then it is also surjective.

T/F 5. If a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is surjective, then its matrix representation in the standard basis is invertible.

Part 2 – Computational exercises

Exercise 1 (Matrix representation, Basis change and similarity). Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ be the linear transformation defined by:

$$T(x_1, x_2, x_3) = (x_1 + 2x_2, x_2 - x_3, 3x_1 + x_3, 2x_1 - x_2 + x_3).$$

Let B be the standard basis of $\mathbb{R}^3$ and C be the standard basis of $\mathbb{R}^4$.

1. Find the matrix representation of T in the standard basis of $\mathbb{R}^3$ and $\mathbb{R}^4$.
2. Justify that the matrix representation of T is unique with respect to these bases.
3. Compute the rank and kernel of the matrix representation of T .
4. Determine if T is injective, surjective, or bijective.
5. Let $B' = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ be a basis of $\mathbb{R}^3$. Find the matrix representation of T in the basis B' and C .
6. Find the change of basis matrix from B to B' . Verify that the matrix representation of T in the new bases is related to the original matrix representation by the change of basis formula.
7. Are the matrix representations of T in the different bases *similar*? Justify your answer.

Exercise 2 (Rank of a matrix). Let $A = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ 1 & 1 & 0 \end{pmatrix}$.

1. Compute the rank of A using row reduction.
2. Find a basis for the column space of A .
3. Find a basis for the null space of A .